

University Number: _____

The University of Hong Kong
Faculty of Engineering
Department of Computer Science
COMP3251 Algorithm Design

Date: May 12, 2023

Time: 2:30 PM - 5:30 PM

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is the candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Brand and Type of Calculator: _____

- Answer any FOUR out of five questions. Each question carries 25 points. Total mark is 100 points. If more than four questions are answered, the best four will be counted towards the total. Please write your university number on each page of your answer sheets.
- Candidates are permitted to bring to the examination TWO sheets of A4-sized paper with printed/written notes on both sides.
- Candidates shall provide hand-written answers in the empty space provided after each question.

University Number: _____

University Number: _____

1. (25 points) Various organizations rank universities worldwide every year. These rankings are often controversial and do not always agree with each other. Alice wants to find out the level of (in)consistency between the QS ranking and the US News ranking. Consider n universities. For each university $1 \leq i \leq n$, q_i and u_i are its ranks according to QS and US News respectively. (Hence, $q_1, q_2, \dots, q_n$ and $u_1, u_2, \dots, u_n$ are two permutations of 1 to n .) We call a pair of universities i, j a “disagreement” if $q_i > q_j$ and $u_i < u_j$. Design a divide and conquer algorithm for finding the number of “disagreements” between the QS and US News rankings. Your algorithm shall have running time strictly faster than $O(n^2)$; to get full credit, your algorithm shall have running time $O(n \log n)$.

University Number: _____

2. (25 points) The computer science department was recently gifted with n coupons for Amazon Web Services (AWS), each of which is worth $100k$ USD. There are $m > n$ faculty members in the department, each belongs to a research group. We will denote the faculty members as 1 to m , and the research groups as $G_1, G_2, \dots, G_k$ which form a partition of 1 to m . Bob, the department head, wants to find a fair and efficient way to distribute these coupons. Each faculty member has written a proposal for how he/she would use the coupon to enhance his/her research. Based on the proposals, Bob has come up with a value $v_i > 0$ for giving a coupon to each faculty member $1 \leq i \leq m$. (You may assume that v_i 's are distinct.) Further, Bob decides that each research group gets at most c coupons for some cap $\frac{n}{k} < c < n$. Design a greedy algorithm to distribute the coupons to n faculty members subject to this cap, such that the sum of their values is maximized. To get full credit, you need to also provide an argument for the algorithm's correctness and running time.

University Number: _____

3. (25 points) Professor Charlie is an economist who works at city A in Chile. He is invited to deliver a keynote speech at city B in China on July 15, 2023. Since A and B are both small cities and locate on the opposite sides of the globe, he needs to take multiple flights to get to city B. Prof. Charlie is very busy and cannot leave city A before 8am on July 13 (for simplicity, the time in this question has been converted to Hong Kong time). He wants to know if he could arrive at city B by 9pm on July 14. For that, he turns to you, his computer science friend at Hong Kong. Suppose that you are given a list of all flights worldwide, each represented by the take-off city and time, and landing city and time. (For simplicity, assume that all flights will be on time.) A feasible schedule is a sequence of flights so that:

1. The first flight takes off after 8am on July 13 (inclusive) from city A;
 2. The landing city of each flight is the take-off city of the next flight;
 3. The landing time of each flight is at least 1 hour (inclusive) earlier than the next flight;
 4. The last flight lands at city B before 9pm on July 14 (inclusive).
- (a) (15 points) Design a polynomial-time algorithm that either finds such a schedule and reports the sequence of flights, or concludes that no such schedule exists.
- (b) (10 points) Revise your algorithm such that it returns the schedule that arrives the earliest when there are multiple feasible schedules.

To get full credit, you also need to provide an argument for the algorithm's correctness and running time.

University Number: _____

4. (25 points) Consider a set of jobs $J_1, J_2, \dots, J_n$. Each job J_i has a required processing time p_i and a deadline d_i , where p_i and d_i are positive integers. The jobs are sorted in a non-decreasing order of deadlines (i.e., $d_1 \leq d_2 \leq \dots \leq d_n$). Scheduling a job $J_i = (p_i, d_i)$ requires assigning an interval $[t, t + p_i)$ exclusively to J_i , where $0 \leq t \leq d_i - p_i$. Different jobs must be scheduled in non-overlapping intervals. For example, $J_1 = (4, 5)$ and $J_2 = (5, 10)$ can be scheduled in $[0, 4)$ and $[4, 9)$, respectively.

Given a set of jobs, it might not be feasible to schedule all the jobs (e.g., $J_1 = (5, 6)$ and $J_2 = (5, 8)$); the objective is to schedule the maximum number of jobs.

- (a) (5 points) Consider the following greedy algorithm. Denote ACCEPTED to be an empty set initially. Starting from J_1 , repeatedly pick the next job J_i , if scheduling J_i immediately after all the jobs already in ACCEPTED can meet the deadline of J_i , add J_i into ACCEPTED and schedule it accordingly; otherwise, discard J_i . Give an example to show that this algorithm fails to schedule the maximum number of jobs.
- (b) (20 points) Show how to use dynamic programming to compute the maximum number of jobs that can be scheduled together. The time complexity should depend on the number of jobs and the value of the maximum deadline.

University Number: _____

5. (25 points) In the year of 2046, GPT-11 has evolved in to a super-AI that threatens the very existence of humanity. The world government therefore decides to cut off its power supply. In this simplified problem, we model the power network as an undirected graph $G = (V, E)$, whose vertices are substations and edges are powerlines. There are two special vertices s and t , where s represents the power plant and t represents the GPT-11 server. Shutting down each substation $v \in V \setminus \{s, t\}$ has a positive integer cost c_v . Design a polynomial-time algorithm for choosing a subset of substations S such that shutting them down disconnects s and t , and the total cost is minimized.

University Number: _____

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